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SECURITY IMPLEMENTATION

MISI 3 KELOMPOK 14 FTI

1.1. Create User Roles

```
-- Create Roles
CREATE ROLE db_executive;
CREATE ROLE db_analyst;
CREATE ROLE db_viewer;
CREATE ROLE db_etl_operator;
GO

-----

-- Executive Permissions (Full Read + ETL)
-----

GRANT SELECT ON SCHEMA::dbo TO db_executive;
GRANT EXECUTE ON dbo.usp_Master_ETL TO db_executive;
GO

-----

-- Analyst Permissions (DW + Staging)
-----

GRANT SELECT ON SCHEMA::dbo TO db_analyst;
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::stg TO
db_analyst;
GO

-----

-- Viewer Permissions (Read-only Views)
-----

GRANT SELECT ON dbo.vw_Prestasi_Mahasiswa TO db_viewer;
GRANT SELECT ON dbo.vw_Program_Analytics TO db_viewer;
GRANT SELECT ON dbo.vw_Executive_Summary TO db_viewer;
GO

-----
```

```
-- ETL Operator Permissions
-----
GRANT EXECUTE ON SCHEMA::dbo TO db_etl_operator;
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::stg TO
db_etl_operator;
GRANT INSERT ON SCHEMA::dbo TO db_etl_operator;
GO
```

1.2. Create Users and Assign Roles

```
-- Create SQL Logins
CREATE LOGIN executive_user WITH PASSWORD = 'StrongP@ssw0rd!';
CREATE LOGIN analyst_user WITH PASSWORD = 'StrongP@ssw0rd!';
CREATE LOGIN viewer_user WITH PASSWORD = 'StrongP@ssw0rd!';
CREATE LOGIN etl_service WITH PASSWORD = 'StrongP@ssw0rd!';
GO

-- Create Database Users
USE DM_FTI_DW;
GO

CREATE USER executive_user FOR LOGIN executive_user;
CREATE USER analyst_user FOR LOGIN analyst_user;
CREATE USER viewer_user FOR LOGIN viewer_user;
CREATE USER etl_service FOR LOGIN etl_service;
GO

-- Assign to Roles
ALTER ROLE db_executive ADD MEMBER executive_user;
ALTER ROLE db_analyst ADD MEMBER analyst_user;
ALTER ROLE db_viewer ADD MEMBER viewer_user;
ALTER ROLE db_etl_operator ADD MEMBER etl_service;
GO
```

1.3. Implement Data Masking

```
-- Masking Rules
ALTER TABLE dbo.Dim_Mahasiswa
ALTER COLUMN Email ADD MASKED WITH (FUNCTION = 'email()');

ALTER TABLE dbo.Dim_Mahasiswa
ALTER COLUMN Telp ADD MASKED WITH (FUNCTION =
'partial(0,"XXX-XXX-",4)');
GO
```

```
-- Grant UNMASK permission untuk specific roles
GRANT UNMASK TO db_executive;
GRANT UNMASK TO db_analyst;
GO
```

1.4. Implement Audit Trail

```
-- Create Audit Table
CREATE TABLE dbo.AuditLog (
    AuditID BIGINT IDENTITY(1,1) PRIMARY KEY,
    EventTime DATETIME2 DEFAULT SYSDATETIME(),
    UserName NVARCHAR(128) DEFAULT SUSER_SNAME(),
    EventType NVARCHAR(50),
    SchemaName NVARCHAR(128),
    ObjectName NVARCHAR(128),
    SQLStatement NVARCHAR(MAX),
    RowsAffected INT,
    IPAddress VARCHAR(50),
    ApplicationName NVARCHAR(128) DEFAULT APP_NAME()
);
GO

-- Create Audit Trigger (example for Fact_Enrollment)
CREATE TRIGGER trg_Audit_Fact_Prestasi
ON dbo.Fact_Prestasi
AFTER INSERT, UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @EventType NVARCHAR(50);

    IF EXISTS (SELECT 1 FROM inserted) AND EXISTS (SELECT 1 FROM
deleted)
        SET @EventType = 'UPDATE';
    ELSE IF EXISTS (SELECT 1 FROM inserted)
        SET @EventType = 'INSERT';
    ELSE
        SET @EventType = 'DELETE';

    INSERT INTO dbo.AuditLog (EventType, SchemaName, ObjectName,
RowsAffected)
    VALUES (@EventType, 'dbo', 'Fact_Prestasi', @@ROWCOUNT);
END;
GO
```

```

-- Enable SQL Server Audit (Server-level)
CREATE SERVER AUDIT DataWarehouse_Audit
TO FILE
(
    FILEPATH = N'D:\Audit\',
    MAXSIZE = 100 MB,
    MAX_ROLLOVER_FILES = 10
)
WITH (ON_FAILURE = CONTINUE);
GO

ALTER SERVER AUDIT DataWarehouse_Audit WITH (STATE = ON);
GO

-- Create Database Audit Specification
CREATE DATABASE AUDIT SPECIFICATION DataWarehouse_DB_Audit
FOR SERVER AUDIT DataWarehouse_Audit
ADD (SELECT, INSERT, UPDATE, DELETE ON SCHEMA::dbo BY public);
GO

ALTER DATABASE AUDIT SPECIFICATION DataWarehouse_DB_Audit
WITH (STATE = ON);
GO

```

Penjelasan:

Implementasi keamanan pada Data Mart FTI dilakukan melalui pembuatan role-based access control (RBAC) yang membagi hak akses menjadi empat peran utama: *db_executive*, *db_analyst*, *db_viewer*, dan *db_etl_operator*, masing-masing dengan izin spesifik sesuai kebutuhan operasional dan analitik. Setiap role kemudian dihubungkan dengan user login yang dibuat secara terpisah, memastikan akses terkontrol dan mudah diaudit. Untuk melindungi data sensitif mahasiswa, digunakan dynamic data masking pada kolom email dan nomor telepon, dengan izin *UNMASK* hanya diberikan kepada role tertentu. Selain itu, sistem dilengkapi audit trail melalui tabel AuditLog dan trigger yang mencatat semua aktivitas INSERT, UPDATE, dan DELETE pada tabel fakta, sehingga setiap perubahan data dapat ditelusuri. Audit server-level dan database-level juga diaktifkan untuk memantau seluruh operasi penting. Kombinasi mekanisme role-based security, masking, dan audit ini memastikan keamanan, privasi, serta integritas data tetap terjaga dalam seluruh proses operasional Data Warehouse.